

Math 104C Project 2: Numerical Methods for Partial Differential Equations

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Code link: <https://github.com/samanthaliujiajia/math104c-project2>

1 Introduction

This project investigates finite-difference methods for elliptic and parabolic partial differential equations. The experiments compare accuracy, stability, mesh refinement, and computational behavior for the five-point finite-difference method, Forward Difference, Backward Difference, and Crank–Nicolson.

The elliptic experiments use square grids with $h = 1/4$, $h = 1/8$, and $h = 1/16$. The heat equation experiments use multiple choices of h and k , including both stable and unstable Forward-Difference cases. Whenever an exact solution is available, the absolute error is computed at mesh points.

2 Numerical Methods

Throughout the report, h denotes the spatial mesh width and k denotes the time step for parabolic problems. For elliptic problems, grid points are $(x_i, y_j) = (ih, jh)$, and w_{ij} is the numerical approximation to $u(x_i, y_j)$. For heat equation problems, $w_{i,j}$ denotes the approximation to $u(x_i, t_j)$, where $x_i = ih$ and $t_j = jk$. The parameter $\lambda = k/h^2$ is the heat equation mesh ratio that controls the stability of the Forward-Difference method.

For elliptic equations on the unit square, the unknowns are the interior grid values w_{ij} . The boundary values are known from the problem statement and are moved to the right-hand side of the linear system. At each interior point, the five-point stencil approximates

$$u_{xx}(x_i, y_j) + u_{yy}(x_i, y_j) \approx \frac{w_{i+1,j} + w_{i-1,j} + w_{i,j+1} + w_{i,j-1} - 4w_{i,j}}{h^2}.$$

For the Laplace problem this gives

$$w_{i+1,j} + w_{i-1,j} + w_{i,j+1} + w_{i,j-1} - 4w_{ij} = 0,$$

while for the Poisson experiment the right-hand side is $h^2 f(x_i, y_j)$. The coefficient matrix is sparse because each interior point only interacts with its four nearest neighbors.

I solve the elliptic systems in two ways. The sparse direct solver is used as a baseline because it solves the matrix equation $Aw = b$ in one algebraic step. Here A is the sparse coefficient matrix

created by the five-point stencil, w is the vector of all unknown interior grid values, and b contains the source term and boundary contributions. I also use successive over-relaxation (SOR), which updates the approximate solution point by point:

$$w_{ij}^{(m+1)} = (1 - \omega)w_{ij}^{(m)} + \frac{\omega}{4} (w_{i+1,j} + w_{i-1,j} + w_{i,j+1} + w_{i,j-1} - h^2 f_{ij}).$$

In this formula, m is the iteration number, $f_{ij} = f(x_i, y_j)$, and ω is the relaxation parameter. In the experiments I use $\omega = 1.6$ and stop when the residual $\|Aw - b\|_\infty$ is below the chosen tolerance. Reporting the iteration count helps compare accuracy with computational effort.

For the heat equation $u_t = u_{xx}$, the Forward-Difference method is

$$w_{i,j+1} = \lambda w_{i-1,j} + (1 - 2\lambda)w_{i,j} + \lambda w_{i+1,j}, \quad \lambda = \frac{k}{h^2}.$$

This method is explicit, so each new time level is computed directly from the previous one. Its simplicity is useful, but it is only conditionally stable for $\lambda \leq 1/2$.

The Backward-Difference method uses the spatial second difference at the new time level:

$$-\lambda w_{i-1,j+1} + (1 + 2\lambda)w_{i,j+1} - \lambda w_{i+1,j+1} = w_{i,j}.$$

This produces a tridiagonal linear system at each time step. It is first-order accurate in time, but its implicit form is unconditionally stable for the heat equation.

Crank–Nicolson averages the spatial second difference between the old and new time levels. In matrix form it can be written as

$$\left(I - \frac{\lambda}{2}A\right) w^{j+1} = \left(I + \frac{\lambda}{2}A\right) w^j,$$

where A is the centered second-difference matrix. This method is also implicit, but it has second-order accuracy in time. Comparing these three heat methods makes the tradeoff between explicit simplicity, implicit stability, and temporal accuracy visible in the numerical results.

Thus the four required methods from the project instructions are all included: the five-point finite-difference method for elliptic PDEs, the Forward-Difference method for the heat equation, the Backward-Difference method for the heat equation, and the Crank–Nicolson method.

3 Problem A: Elliptic PDE

Problem A solves

$$u_{xx} + u_{yy} = 0, \quad 0 < x < 1, \quad 0 < y < 1,$$

with boundary conditions

$$u(0, y) = 0, \quad u(x, 0) = 0, \quad u(x, 1) = 100x, \quad u(1, y) = 100y.$$

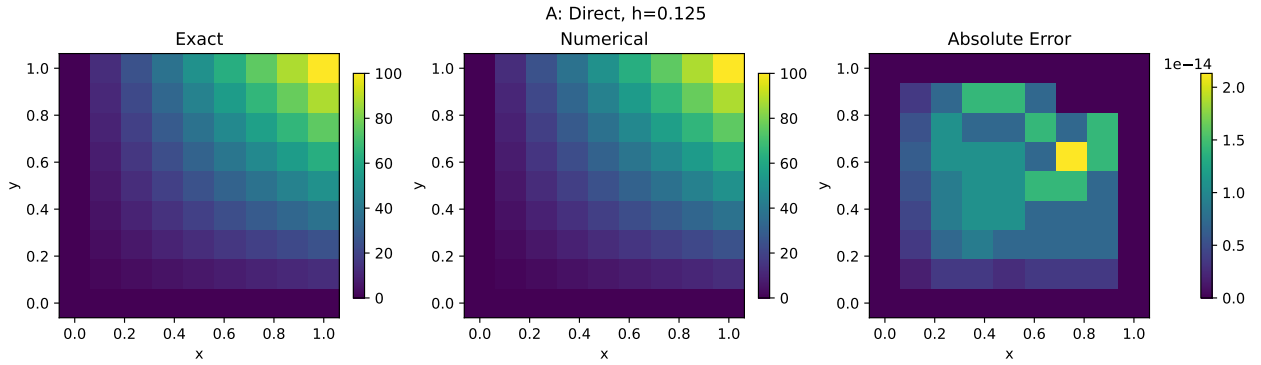
The exact solution is $u(x, y) = 100xy$, which is bilinear. Since the five-point stencil is exact for this function up to roundoff, this problem is a useful check that boundary values and matrix assembly are correct.

Table 1: Elliptic PDE error summary and SOR iteration counts.

problem	method	h	max error	iterations	residual
A	Direct	0.250000	7.105e-15	0	2.842e-14
A	SOR omega=1.6	0.250000	5.321e-11	56	1.873e-10
A	Direct	0.125000	2.132e-14	0	3.553e-14
A	SOR omega=1.6	0.125000	6.425e-11	58	1.914e-10
A	Direct	0.062500	2.132e-14	0	8.527e-14
A	SOR omega=1.6	0.062500	8.285e-10	137	9.442e-11
C1	Direct	0.250000	0.053029	0	8.882e-16
C1	SOR omega=1.6	0.250000	0.053029	47	1.591e-10
C1	Direct	0.125000	0.012951	0	6.384e-16
C1	SOR omega=1.6	0.125000	0.012951	48	8.876e-11
C1	Direct	0.062500	0.003219	0	1.527e-15
C1	SOR omega=1.6	0.062500	0.003219	114	8.457e-11

Table 2: Observed orders for elliptic experiments.

problem	method	observed order
A	Direct	-0.792481
A	SOR omega=1.6	-1.980335
C1	Direct	2.021060
C1	SOR omega=1.6	2.021061

Figure 1: Problem A exact solution, numerical solution, and absolute error for $h = 1/8$.

4 Problem B: Heat Equation

Problem B solves

$$u_t = u_{xx}, \quad 0 < x < 1, \quad t > 0,$$

with $u(0, t) = u(1, t) = 0$ and $u(x, 0) = \sin(\pi x)$. The exact solution is

$$u(x, t) = e^{-\pi^2 t} \sin(\pi x).$$

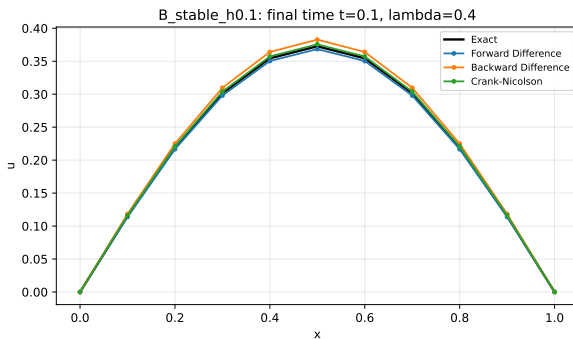
The Forward-Difference experiments include $\lambda = 0.4$, which satisfies the stability condition, and $\lambda = 0.6$, which violates it.

Table 3: Heat equation error summary.

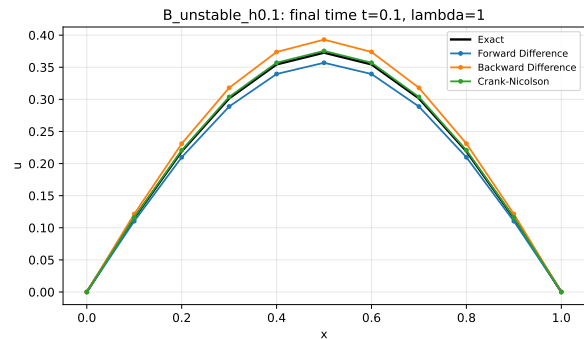
problem	case	method	h	k	λ	final error	max error
B	B_stable_h0.1	Forward Difference	0.100000	0.004000	0.400000	0.004294	0.004294
B	B_stable_h0.1	Backward Difference	0.100000	0.004000	0.400000	0.010112	0.010112
B	B_stable_h0.1	Crank-Nicolson	0.100000	0.004000	0.400000	0.002981	0.002981
B	B_refined_h0.05	Forward Difference	0.050000	0.001000	0.400000	0.001063	0.001063
B	B_refined_h0.05	Backward Difference	0.050000	0.001000	0.400000	0.002561	0.002561
B	B_refined_h0.05	Crank-Nicolson	0.050000	0.001000	0.400000	7.535e-04	7.535e-04
B	B_unstable_h0.1	Forward Difference	0.100000	0.010000	1.000000	0.015756	0.015756
B	B_unstable_h0.1	Backward Difference	0.100000	0.010000	1.000000	0.020320	0.020320
B	B_unstable_h0.1	Crank-Nicolson	0.100000	0.010000	1.000000	0.002734	0.002734
C2	C2_stable_h0.1	Forward Difference	0.100000	0.004000	0.400000	0.003285	0.017047
C2	C2_stable_h0.1	Backward Difference	0.100000	0.004000	0.400000	0.008843	0.036656
C2	C2_stable_h0.1	Crank-Nicolson	0.100000	0.004000	0.400000	0.002355	0.010840
C2	C2_refined_h0.05	Forward Difference	0.050000	0.001000	0.400000	8.740e-04	0.004294
C2	C2_refined_h0.05	Backward Difference	0.050000	0.001000	0.400000	0.002181	0.010114
C2	C2_refined_h0.05	Crank-Nicolson	0.050000	0.001000	0.400000	6.247e-04	0.002981

Table 4: Observed orders for stable heat equation refinement cases.

problem	method	observed order
B	Backward Difference	1.981501
B	Crank-Nicolson	1.983931
B	Forward Difference	2.014890
C2	Backward Difference	1.857639
C2	Crank-Nicolson	1.862650
C2	Forward Difference	1.989060



Stable Forward case, $\lambda = 0.4$.



Unstable Forward case, $\lambda = 0.6$.

Figure 2: Problem B final-time solution profiles.

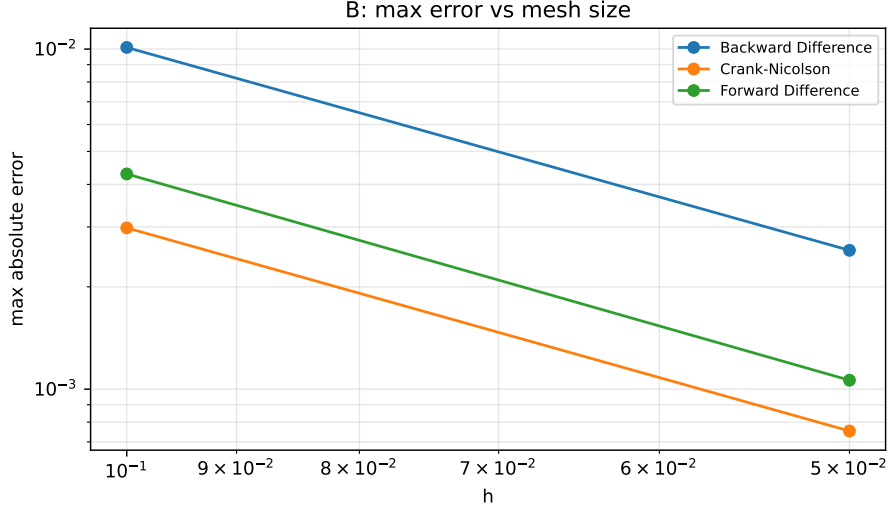


Figure 3: Problem B max error under stable mesh refinement.

5 Problem C: Additional Experiments

For Problem C, I chose two of the listed additional experiment options: **Option 1(b)**, adding a source term and solving a Poisson equation, and **Option 2**, changing the initial condition for the heat equation. These two choices extend the required Problems A and B without changing the overall finite-difference framework, so the results can be compared directly with the earlier experiments.

Experiment C1: Poisson Equation with a Source Term

This experiment investigates the Poisson problem

$$u_{xx} + u_{yy} = -2\pi^2 \sin(\pi x) \sin(\pi y), \quad u = 0 \text{ on the boundary.}$$

This corresponds to option 1(b) from the project instructions. The source term is chosen by the method of manufactured solutions so that the exact solution is

$$u(x, y) = \sin(\pi x) \sin(\pi y).$$

This experiment is interesting because it keeps the same elliptic five-point stencil as Problem A, but it removes the special bilinear exactness of $u(x, y) = 100xy$. In Problem A, the stencil reproduces the exact solution up to roundoff error. Here, the solution has curvature in both spatial directions, so the $O(h^2)$ truncation error of the five-point method becomes visible.

The numerical results are shown in the elliptic error table and in the figures below. The heatmap for $h = 1/8$ compares the exact solution, the numerical solution, and the absolute error. The convergence plot shows that the error decreases as h is refined from $1/4$ to $1/16$. This behavior agrees with the theory for the centered five-point approximation, whose local truncation error is second order in space.

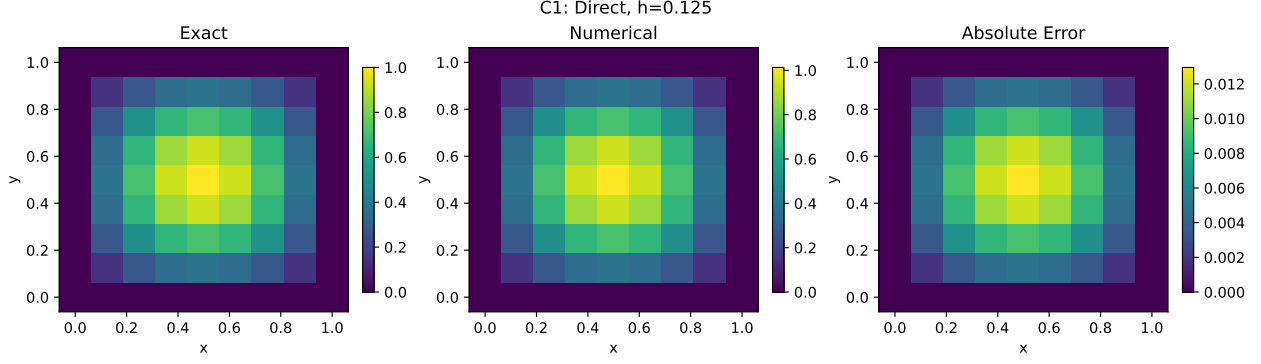


Figure 4: Poisson manufactured-solution experiment for $h = 1/8$.

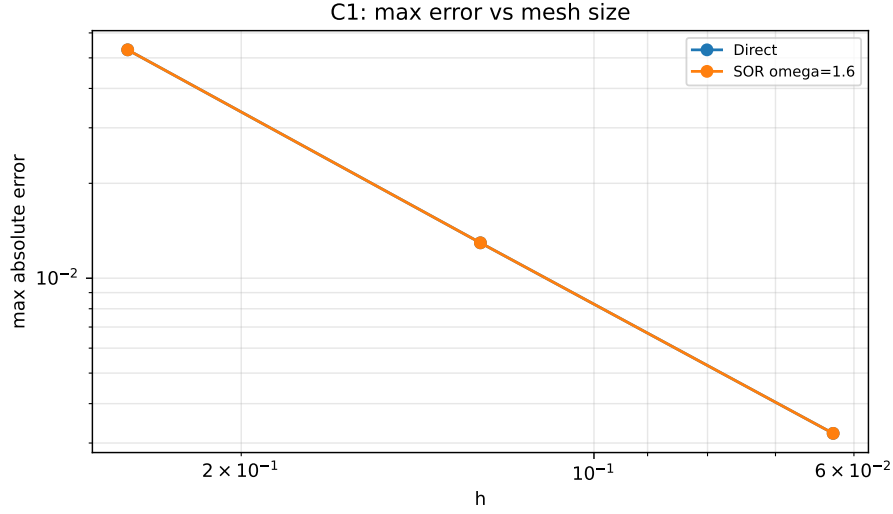


Figure 5: Problem C1 max error under mesh refinement.

Experiment C2: Different Heat Equation Initial Condition

This experiment investigates option 2 from the project instructions: changing the initial condition for the heat equation. Instead of using $\sin(\pi x)$, as in Problem B, I use

$$u(x, 0) = \sin(2\pi x),$$

whose exact solution is

$$u(x, t) = e^{-4\pi^2 t} \sin(2\pi x).$$

The same boundary conditions $u(0, t) = u(1, t) = 0$ are used.

This experiment is interesting because $\sin(2\pi x)$ is a higher-frequency mode than the initial condition in Problem B. It oscillates more rapidly in space and decays faster in time, since the exponential factor is $e^{-4\pi^2 t}$ instead of $e^{-\pi^2 t}$. A coarse grid has fewer points per wavelength, so the spatial discretization has a harder job resolving the solution accurately.

The numerical results are shown in the heat equation error table and in the figures below. The final-time profile compares Forward Difference, Backward Difference, and Crank–Nicolson against the exact solution. The convergence plot shows how the error changes under stable mesh refinement. The observed behavior is consistent with heat equation theory: physical high-frequency modes decay quickly, but numerical accuracy still depends on resolving the spatial mode and choosing a stable time step.

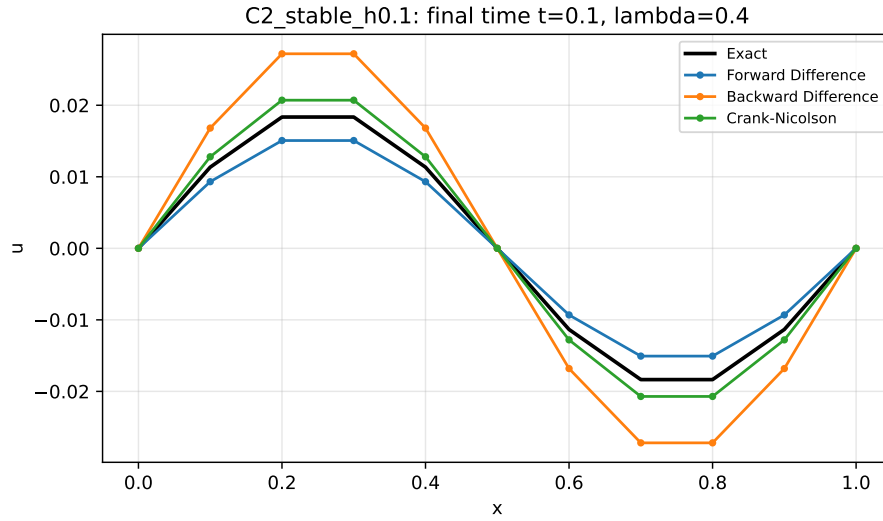


Figure 6: Problem C2 final-time profiles for the different initial condition.

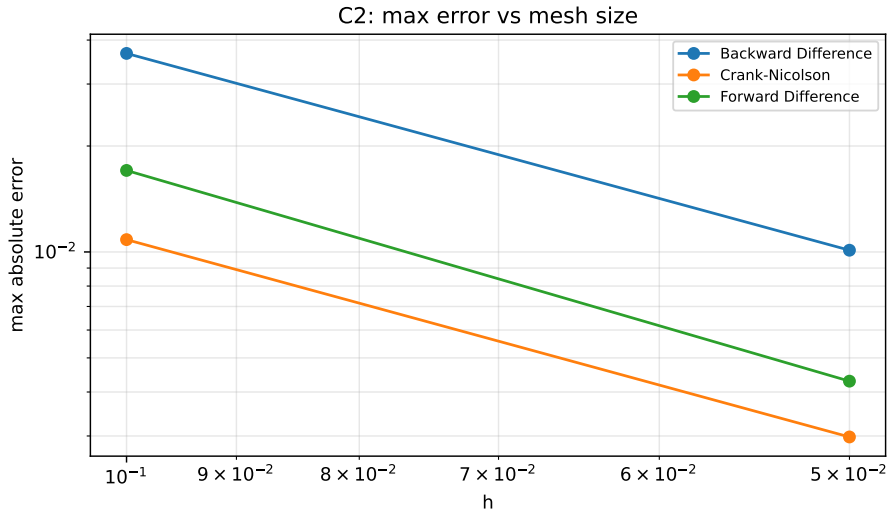


Figure 7: Problem C2 max error under stable mesh refinement.

6 Discussion and Analysis

Problem A confirms both the finite-difference stencil and the treatment of boundary data. Since the exact solution $u(x, y) = 100xy$ is bilinear, the second derivatives are represented exactly by the centered difference formula up to roundoff error. Therefore, the small errors in the table are not surprising: they mainly measure floating-point error and the accuracy of the linear solve, not truncation error. This is still an important test because an incorrect sign in the matrix or a missing boundary contribution would immediately destroy the linear solution.

The direct solver and SOR solver give essentially the same numerical solution, but their computational behavior is different. The sparse direct solver treats the linear system as one algebraic problem. SOR updates the grid values repeatedly, so the number of iterations grows as the mesh is refined. This is a typical feature of stationary iterative methods: finer grids have more unknowns, and information moves across the domain gradually through local updates. The residuals show that the iteration was stopped only after the algebraic system was solved to a tight tolerance.

Problem B illustrates the central stability difference between explicit and implicit time stepping. For the heat equation, the Forward-Difference method is consistent with local truncation error $O(k + h^2)$, but consistency alone does not guarantee useful numerical results. The method also needs $\lambda = k/h^2 \leq 1/2$. In the stable case, the final-time profiles remain smooth and the error decreases under refinement. In the unstable case, the method amplifies high-frequency error modes instead of damping them, so the computed solution can develop nonphysical oscillations even though the exact heat equation smooths and decays the solution.

Backward Difference and Crank–Nicolson behave differently because the new time level is included implicitly. Backward Difference is more diffusive, which can make it slightly less accurate for the same mesh, but that diffusion is also why it strongly damps unstable modes. Crank–Nicolson is less artificially dissipative and has second-order temporal accuracy, so it usually gives a better balance between accuracy and stability. The experiments show this tradeoff: implicit methods require solving a linear system at every time step, but they allow larger k without the Forward-Difference stability failure.

The Poisson manufactured-solution experiment is a better test of spatial convergence than Problem A because $u = \sin(\pi x)\sin(\pi y)$ is not exactly represented by the five-point stencil. The observed errors decrease as the mesh is refined, which matches the theoretical second-order behavior of the centered spatial approximation. This experiment also shows why manufactured solutions are useful: the source term is chosen so the exact solution is known, making it possible to measure the numerical error directly.

The second additional experiment changes the heat equation initial condition to $\sin(2\pi x)$. This mode has a higher spatial frequency and decays like $e^{-4\pi^2 t}$, much faster than the $\sin(\pi x)$ mode. Numerically, this means a coarse mesh has fewer points per wavelength, so spatial resolution matters more. The experiment connects error propagation with the structure of the PDE: the heat equation damps high-frequency physical modes, but a numerical method can still introduce additional error if the grid or time step is too coarse.

Overall, the experiments show the relationship between consistency, stability, and convergence. The finite-difference formulas are consistent approximations, but stability determines whether the accumulated error remains controlled. For elliptic problems, the main computational issue is solving the sparse linear system accurately and efficiently. For parabolic problems, the choice of time step

is just as important as the spatial mesh, especially for explicit methods. Thus the best method depends not only on formal order, but also on stability requirements and computational cost.

7 Conclusion

The experiments support the theoretical results for finite-difference PDE methods. The five-point elliptic method is accurate and has a sparse matrix structure that can be solved either directly or iteratively. For parabolic problems, consistency alone is not enough: the Forward-Difference method also requires the stability condition $\lambda \leq 1/2$. Backward Difference and Crank–Nicolson are more stable for large time steps, with Crank–Nicolson offering higher temporal accuracy.